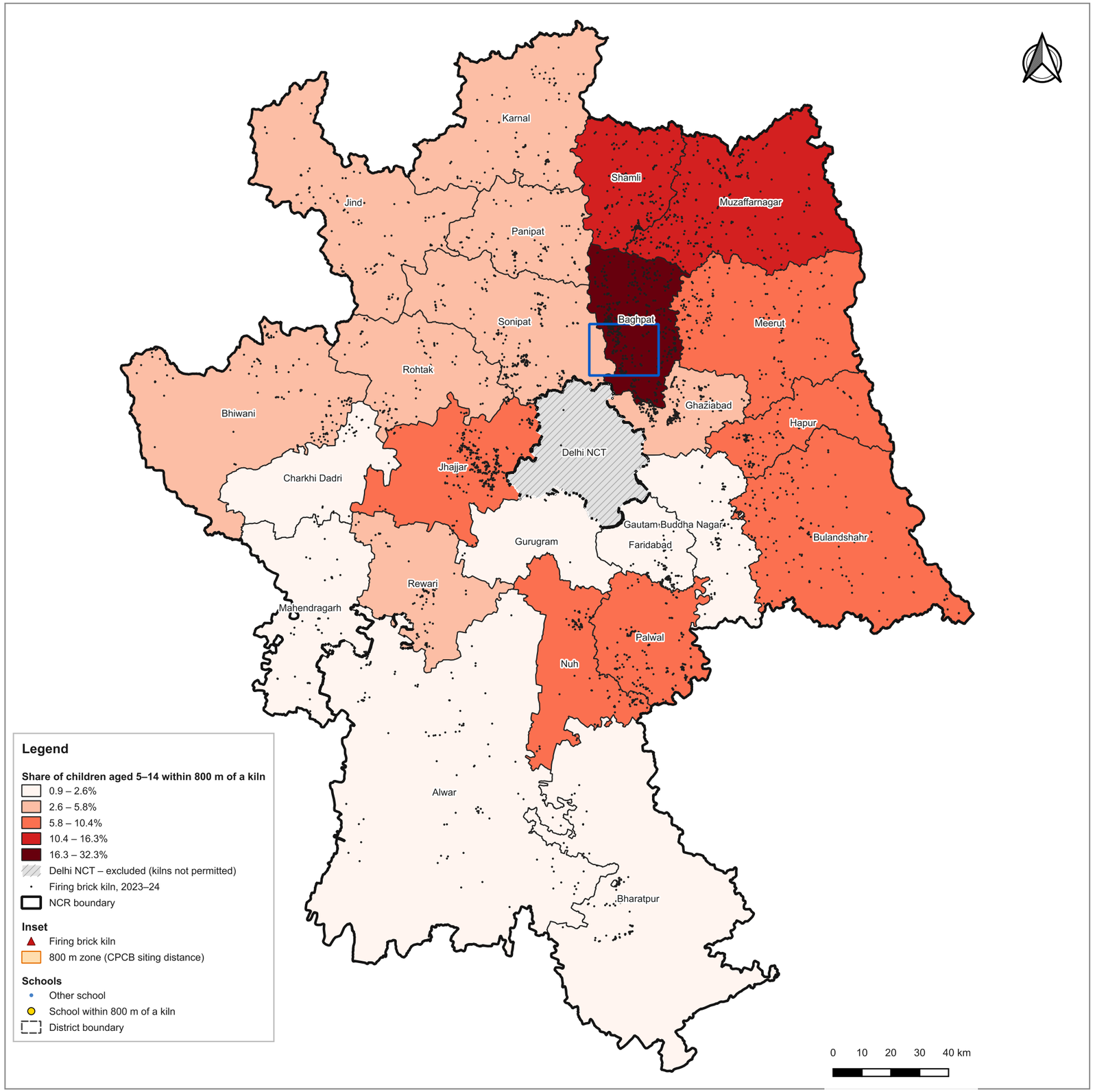
One in three children in Baghpat lives within 800 m of a firing brick kiln

Brick kilns found by AI on Sentinel-2 imagery, and the children and schools inside the CPCB 800 m siting distance · National Capital Region, 2023–24 firing season

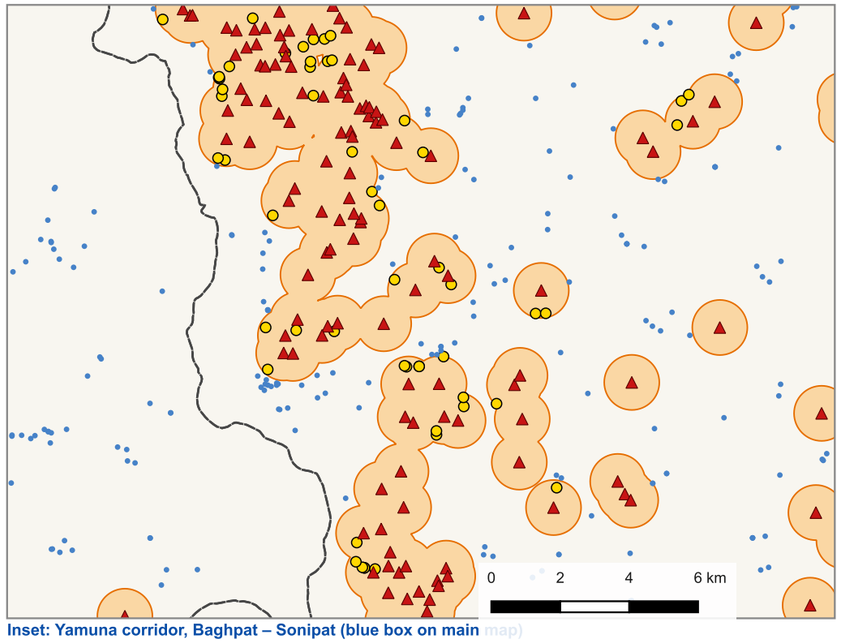


Legend

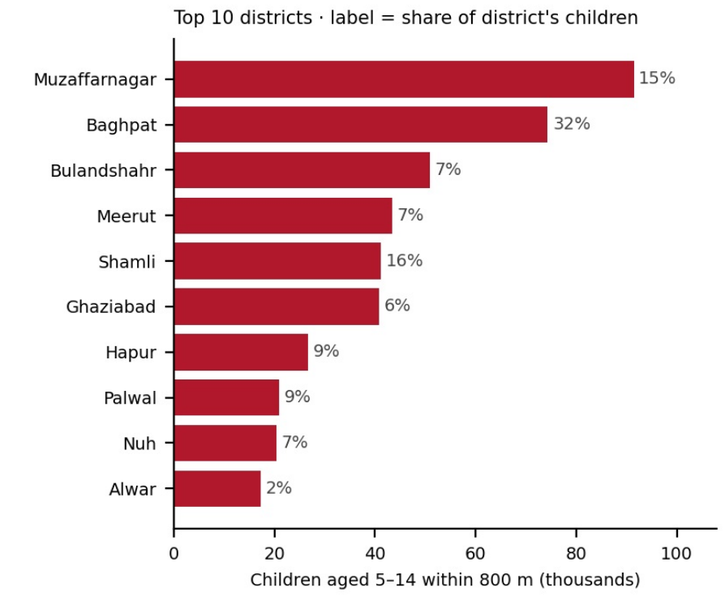
Share of children aged 5–14 within 800 m of a kiln
- 0.9 – 2.6%
- 2.6 – 5.8%
- 5.8 – 10.4%
- 10.4 – 16.3%
- 16.3 – 32.3%
- Delhi NCT – excluded (kilns not permitted)
- Firing brick kiln, 2023–24
- NCR boundary

Inset
- Firing brick kiln
- 800 m zone (CPCB siting distance)

Schools
- Other school
- School within 800 m of a kiln
- District boundary



Inset: Yamuna corridor, Baghpat – Sonipat (blue box on main map)



Key numbers (NCR excluding Delhi)

≈ 539,000 children aged 5–14 (6.5%) live within 800 m of a firing kiln
1,191 schools stand within 800 m · 32% of 3,096 kilns have a school within 800 m
Kilns near schools were no more likely to stop firing by 2025–26 (10.8% vs 10.9%)

Method: YOLO11 oriented-box detector on Sentinel-2 winter composites, filtered by land cover, verified by SWIR hot-spot firing frequency, non-kiln heat sources removed; every kiln near a school checked by eye on Esri imagery and every weak-firing kiln found with Earth Engine (166 non-kilns removed). Children: WorldPop 2020, ages 5–14. Schools: UDISE (DataMeet), location-checked, shared points removed.
Data: Copernicus Sentinel-2 (ESA), SentinelKilnDB (IIT Gandhinagar, CC BY-NC 4.0), ESA WorldCover, WorldPop, UDISE/DataMeet, OpenStreetMap contributors, NCRPB boundaries (rebuilt).
CRS: WGS 84 / UTM zone 43N (EPSG:32643) · Produced 14 October 2026 · KilnWatch – Rashmi Sinha